

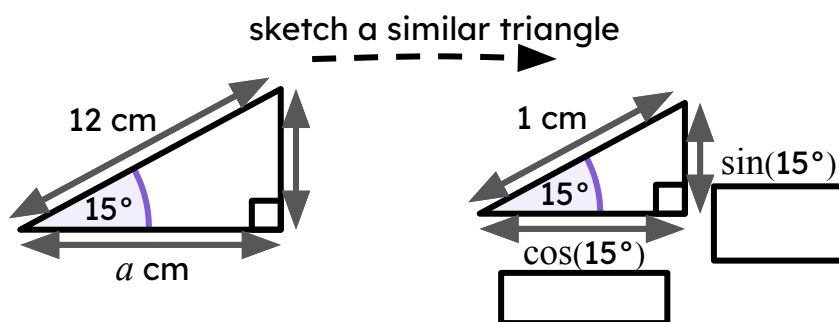


Sine and cosine ratios

Task A: Ratio tables with the unit circle

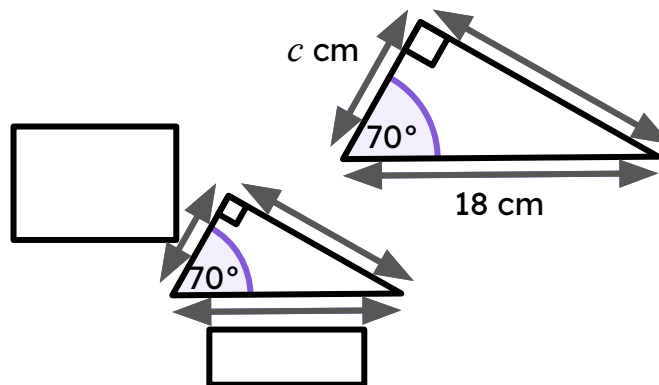
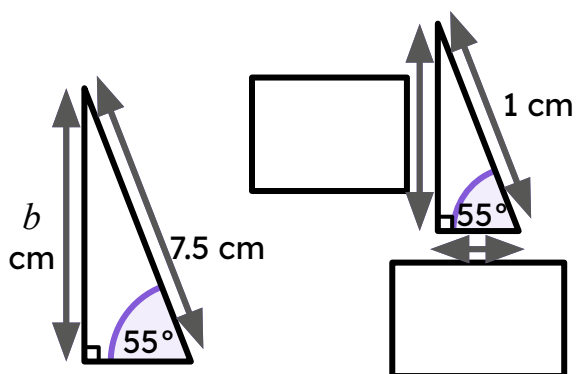
angle	sine	cosine	tangent
0	0	1	0
5	0.087	0.996	0.087
10	0.174	0.985	0.176
15	0.259	0.966	0.268
20	0.342	0.94	0.364
25	0.423	0.906	0.466
30	0.5	0.866	0.577
35	0.574	0.819	0.7
40	0.643	0.766	0.839
45	0.707	0.707	1
50	0.766	0.643	1.192
55	0.819	0.574	1.428
60	0.866	0.5	1.732
65	0.906	0.423	2.145
70	0.94	0.342	2.747
75	0.966	0.259	3.732
80	0.985	0.174	5.671
85	0.996	0.087	11.43
90	1	0	∞

1) Fill in the blanks and ratio table. By considering corresponding sides, find the length of side labelled a cm.



	hypotenuse	
	1	
	12	

2) Fill in all the blanks and the ratio tables to find the values of b and c by considering corresponding sides.



	hypotenuse	
	1	
	7.5	

	hypotenuse	
	1	
	18	

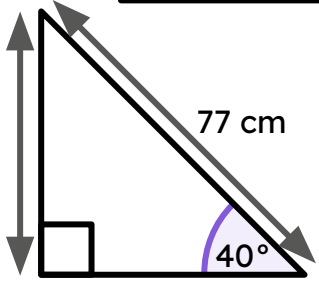
Task B: The opposite and the hypotenuse

1) Find the length of the side opposite the labelled angle using this relationship.

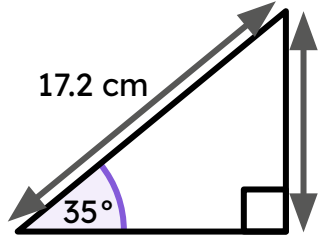
$$\text{opposite side} = h \times \sin(\theta^\circ)$$

a)

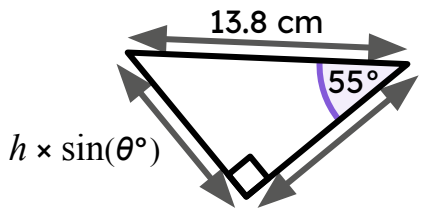
$h \times \sin(\theta^\circ) \text{ cm}$
 $\square \times \sin(\square) \text{ cm}$
 $\square \times \square \text{ cm}$



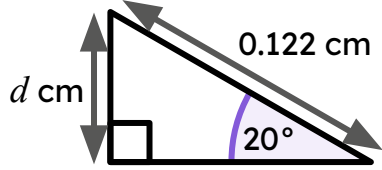
b)



c)



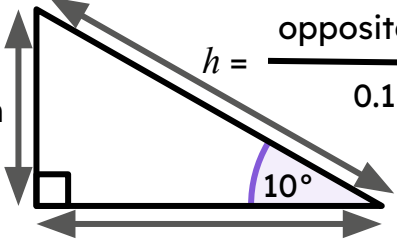
d)



2) Find the length of the hypotenuse when given the side opposite the labelled angle.

$$\text{opposite side} = h \times \sin(\theta^\circ)$$

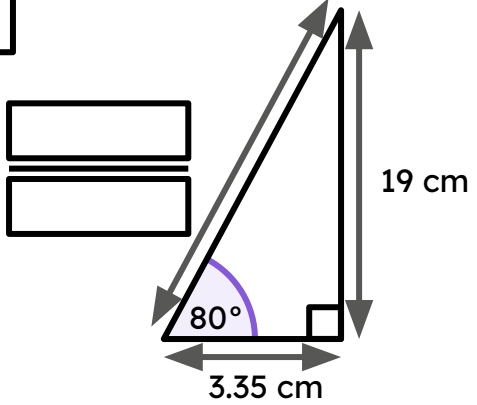
a)



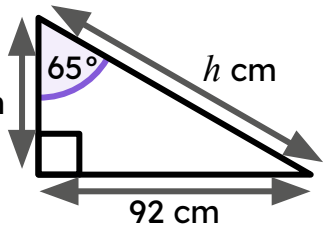
$h = \frac{\text{opposite side}}{0.174}$

b)

$h = \frac{\square}{\square}$



c)



d) A right-angled triangle has an angle of 85° . The length of the side opposite this angle is 142 cm.

Task C: The adjacent and the hypotenuse

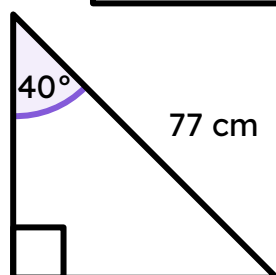
1) Find the length of the side adjacent to the labelled angle using this relationship.

a)

$$h \times \cos(\theta^\circ) \text{ cm}$$

$$\square \times \cos(\square) \text{ cm}$$

$$\square \times \square \text{ cm}$$

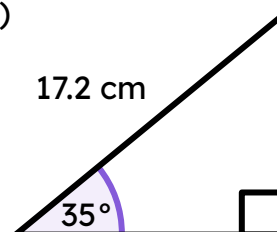


$$\text{adjacent side} = h \times \cos(\theta^\circ)$$

b)

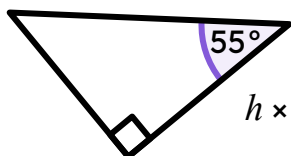
$$17.2 \text{ cm}$$

$$\square \times \cos(\square) \text{ cm}$$



c)

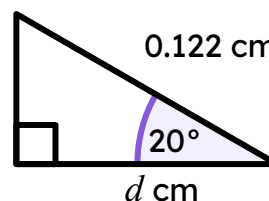
$$13.8 \text{ cm}$$



$$h \times \cos(\theta^\circ)$$

d)

$$0.122 \text{ cm}$$



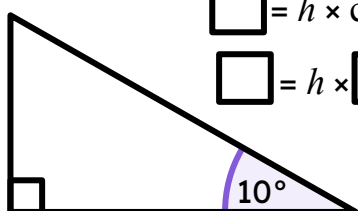
$$d \text{ cm}$$

2) Find the length of the hypotenuse when given the side adjacent to the labelled angle.

a)

$$\square = h \times \cos(\square)$$

$$\square = h \times \square$$

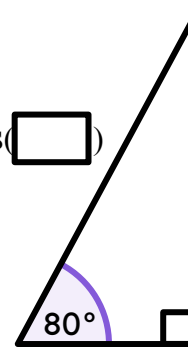


$$18 \text{ cm}$$

$$\text{adjacent side} = h \times \cos(\theta^\circ)$$

b)

$$\square = h \times \cos(\square)$$

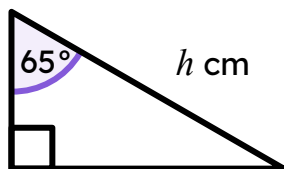


$$19 \text{ cm}$$

$$3.35 \text{ cm}$$

c)

$$43 \text{ cm}$$



$$h \text{ cm}$$

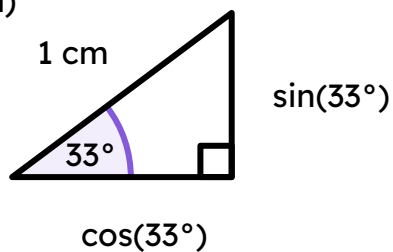
d) A right-angled triangle has an angle of 35° . The length of the side that is adjacent to this angle and the right-angle is 142 cm.



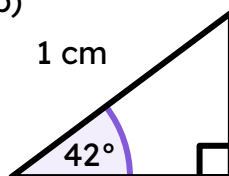
Task D: Using a calculator without a unit circle

1) Find the lengths of all sides in each of these triangles (3 d.p.)

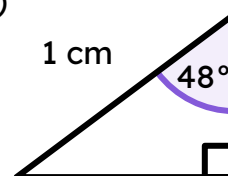
a)



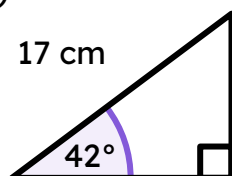
b)



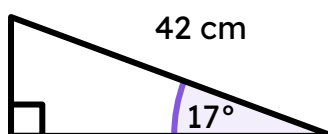
c)



d)



e)



f)

